

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

STUDY MATERIALS



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PART II: BASIC ELECTRONICS ENGINEERING
(2019 Scheme)

Max. Marks: 50

Duration: 90 min

PART A*Answer all questions, each carries 4 marks.*

12. Find the capacitance values for the following codes (i) 2n2 (ii) 104K.
13. What do you mean by majority and minority carriers in a semiconductor?
14. Describe the block diagram of a public addressing system
15. Narrate the working of a capacitor filter.
16. Write the frequency range and typical applications of VHF and UHF (5x4=20) frequency bands.

PART B*Answer one full question from each module, each question carries 10 marks***Module-IV**

- 17 a) Explain the formation of potential barrier in a P-N junction diode. (5)
- b) Draw and explain the V-I characteristics of a PN junction diode under forward and reverse bias. (5)

OR

- 18 a) Explain the working of an NPN transistor mentioning all current components. (6)
- b) The dc current gain of a transistor in common emitter configuration is 100. Find its dc current gain in common base configuration. (4)

Module-V

- 19 a) Sketch the block diagram of a DC power supply and explain the role of each block. (5)
- b) Draw and explain the working of a full wave bridge rectifier circuit. (5)

OR

- 20 a) Sketch the frequency response of a transistor amplifier and comment on the shape of the curve. (7)
- b) What is the role of emitter resistor in an RC coupled amplifier? (3)

Module-VI

- 21 a) State the merits and demerits of Amplitude Modulation. (4)
- b) Sketch the block diagram of a superheterodyne receiver and explain its working. (6)

OR

- 22 a) Describe the working principle of an antenna. (3)
- b) Draw the block diagram of a GSM system and explain its working. (7)

12)

(i) $2n2 = 2 \cdot 2 \text{ nano nF}$

$= 2200 \text{ pF}$

$2n2 = 222$ - common capacitor code

(ii) $104k$

third digit is 4, so four zeros after first two digit

$10000 \text{ pF} = 100 \text{ nF}$

13) * The less abundant charge carriers are called minority carriers.

eg:- In n-type, they are holes

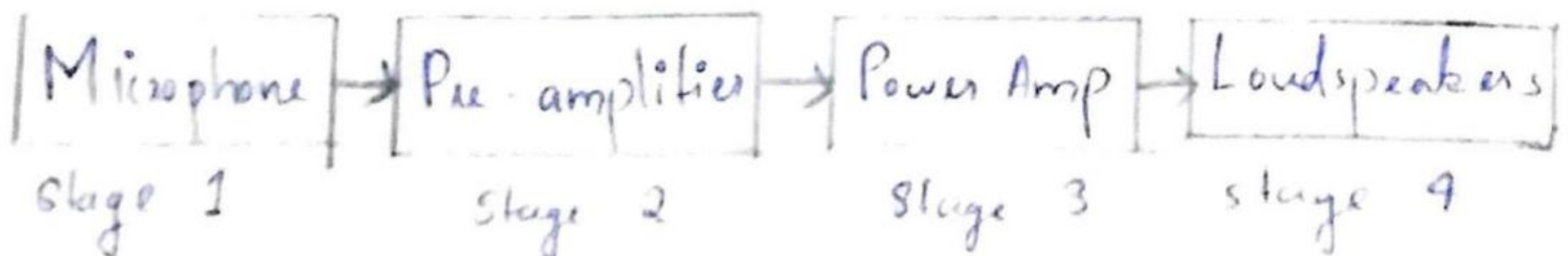
In p-type, they are electrons

* The charge carrier present in large quantity are called majority carriers

eg:- In n-type, they are electrons

In p-type, they are holes

14)



Stage 1 : Microphone (Transducer)

- * Converts sound waves into electrical signals
- * It is important that microphone create a faithful reproduction of sound waves as an electrical signal

Stage 2 : Pre-Amplifier

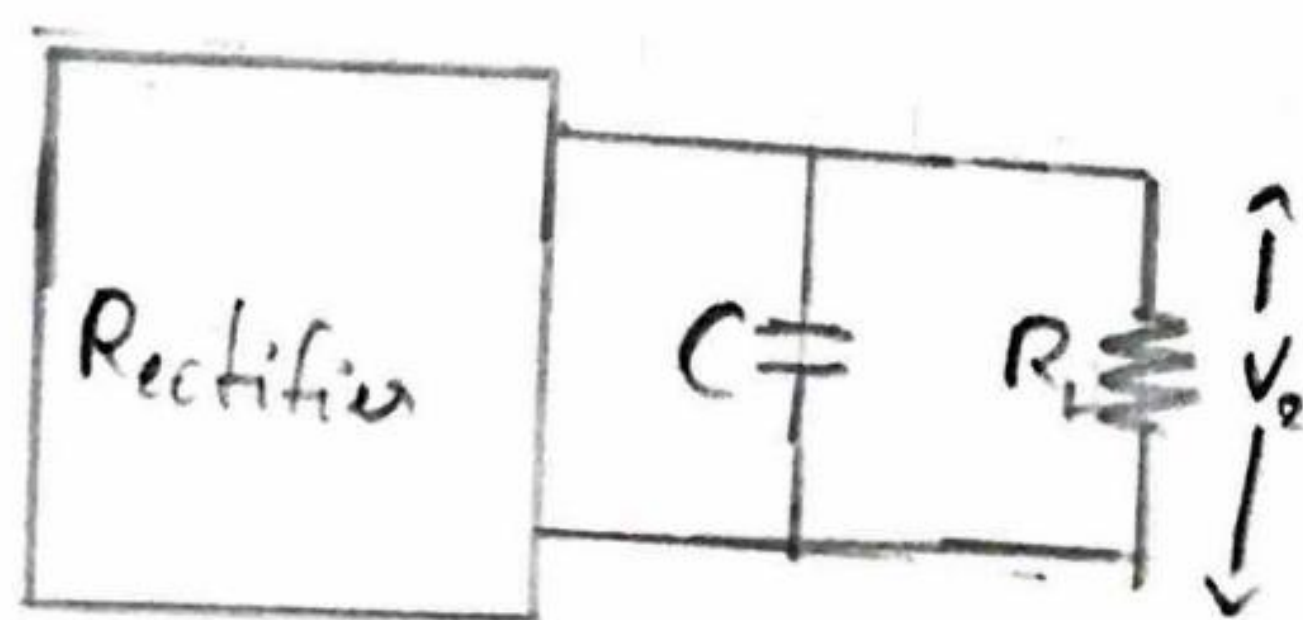
- * Basically a voltage amplifier
- * Its purpose is to take the small electrical signals from the microphone and increase the amplitude of the signal voltage.

Stage 3 : Power amplifier

- * takes this enlarged voltage signal, and boosts the current so that it is strong enough to drive the loudspeaker

Stage 4 : Loud speaker

- * The electrical signal is transformed back into a sound wave



15

- * Connect a large value capacitor C in shunt with the load resistor.
- * When the rectifier output voltage is increasing, the capacitor charges to peak voltage V_m . After
- * After the positive peak is passed, the rectifier output voltage falls which permits the capacitor to discharge through

* To reduce ripple in rectified output, we use the capacitor to discharge slowly.

16) VHF - Very high frequency (30 - 300 MHz)

* Used for TV, FM transmission, radar, telepho

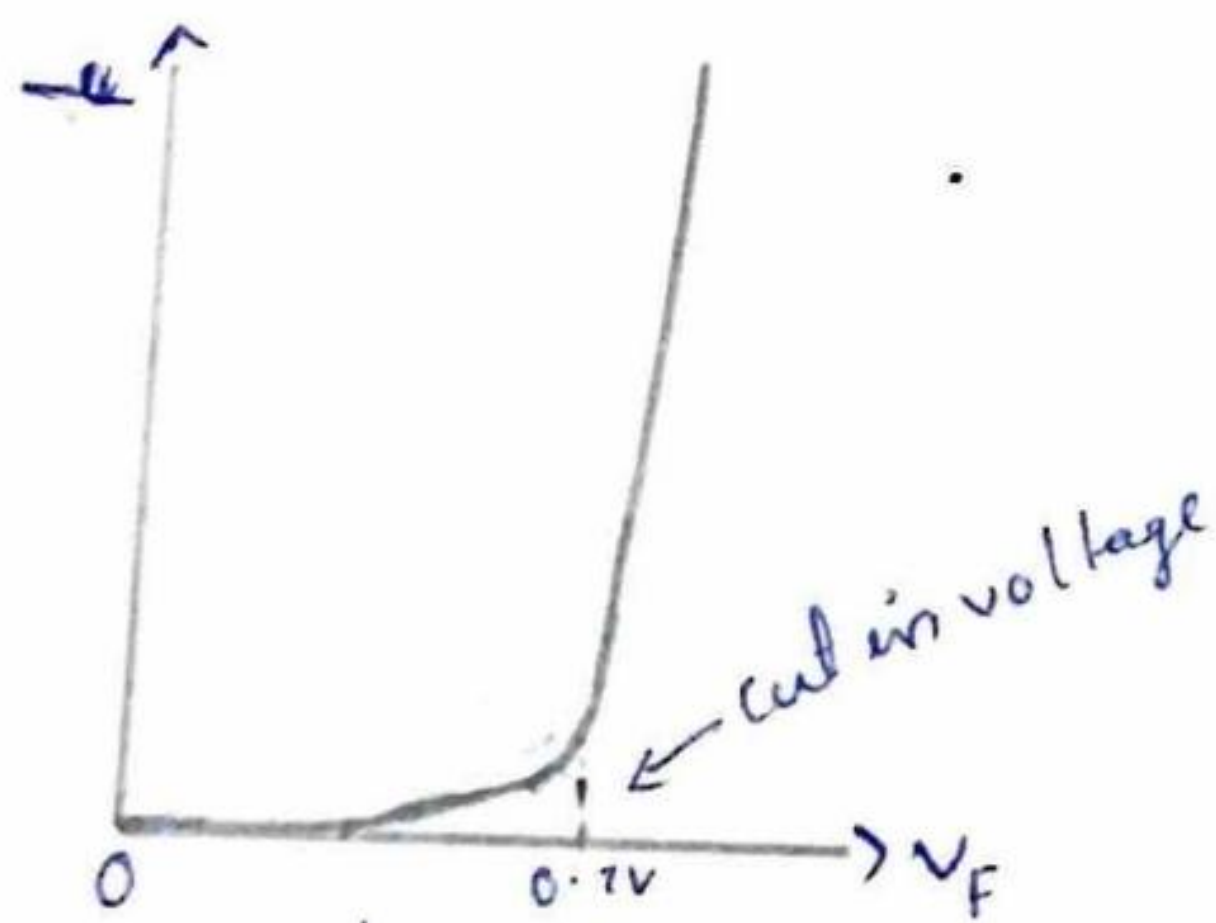
UHF - Ultra high frequency (300 MHz - 3 GHz)

* Used for TV and short distance commu

17 a) When a PN junction is formed, electrons move from n-side to p-side. They will meet at the junction and hole recombination takes place. This will result in the formation of a region near the junction and it is free of electrons and holes. This region is called depletion region. In this region only immobile donor and acceptor ions are present. This will result in a separation of charges. This charge separation results in a potential difference between N-side and P-side. This potential difference further restricts the movement of holes and electrons. Hence this potential is called barrier potential. That region may be called as space-charge region.

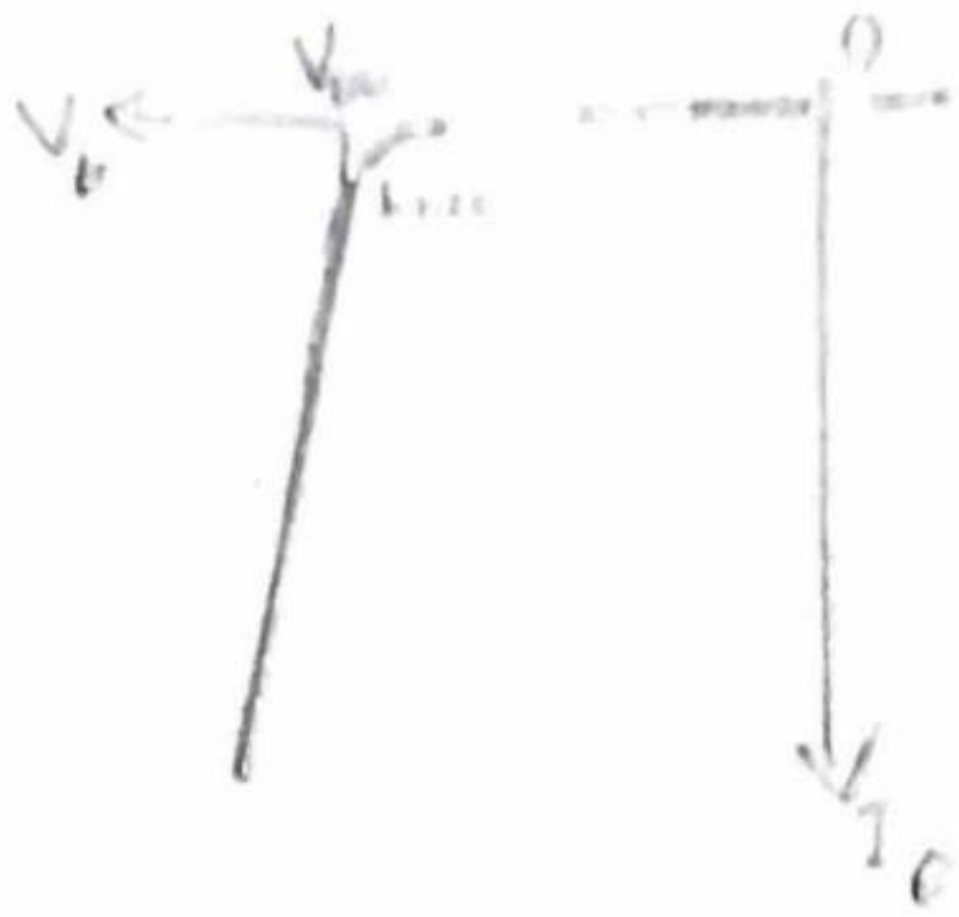
17b) Forward biased V-I characteristics:-

- * The applied voltage (V_F) increases beyond the certain value, the forward current I_F increases sharply
- * The diode permits the current to flow only if external applied voltage overcomes the barrier potential
- * Minimum voltage required for the diode to conduct is called cut-in voltage.
- * If forward voltage is increased beyond a certain safe value, it will produce an extremely large current which may destroy the junction due to overheating



Reverse biased ~~characteristic~~ characteristic

- * The reverse current caused by thermally produced minority carriers is called reverse saturation current or leakage current.
- * If reverse voltage is increased continuously beyond a knee value, breakdown of junction occurs



18a) n-p-n transistor

mainly there are three regions

1. Emitter, E

- * Supplies carrier
- * Heavily doped
- * Emitter region & collector region are same type
- * Medium in size

2. Base, B

- * Lightly doped
- * Smallest region
- * Control the flow of current

3. Collector, C

- * Collect carrier
- * Moderately doped
- * Largest region

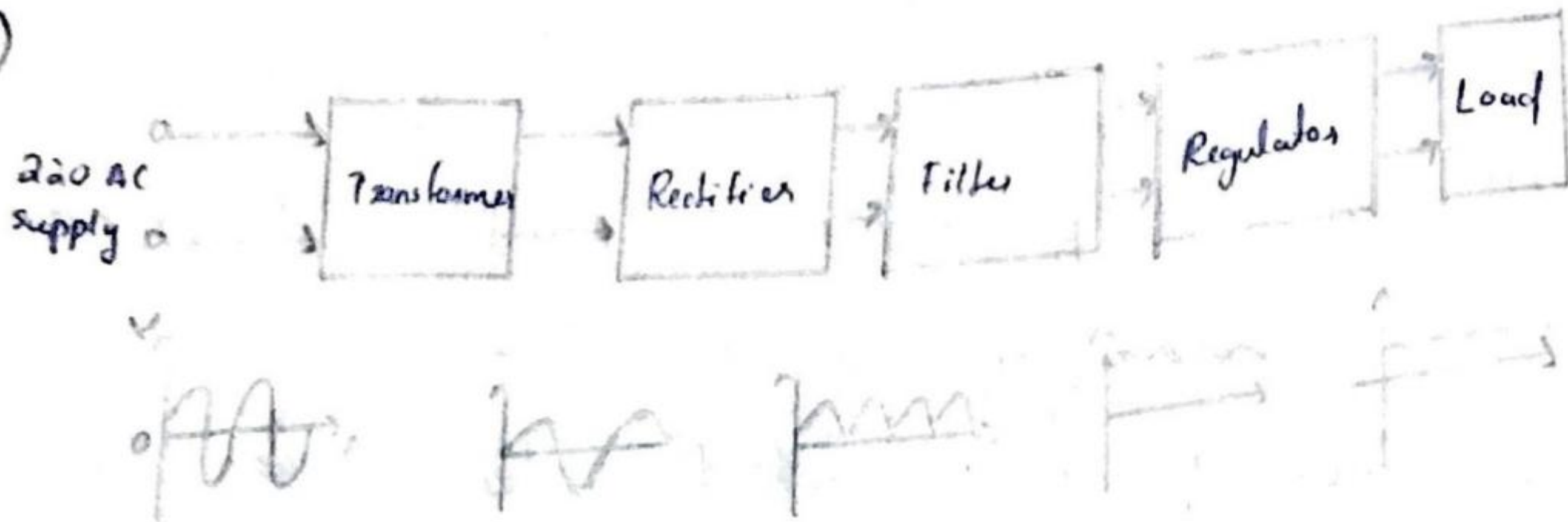
Working:-

Part (b) NPN Transistor,

The forward biased is applied across the emitter-base junction, and the reversed bias is applied across the ~~emitter~~^{collector} base junction. The forward biased voltage V_{EB} is small as compared to the reverse bias voltage V_{CB} . The emitter of NPN transistor is heavily doped. When the forward bias is applied across the emitter, the majority charge carriers move towards the base. This causes the emitter current I_E . The e^- enter into the p-type material and combine with the holes. Base is lightly doped. So only few electrons are combined and remaining constitute the base current I_B . This base current enter into the collector region. The reverse bias potential of the collector region applies the high attractive force on the electrons ~~are~~ reaching collector junction. Thus attract or collect the electrons at the collector,

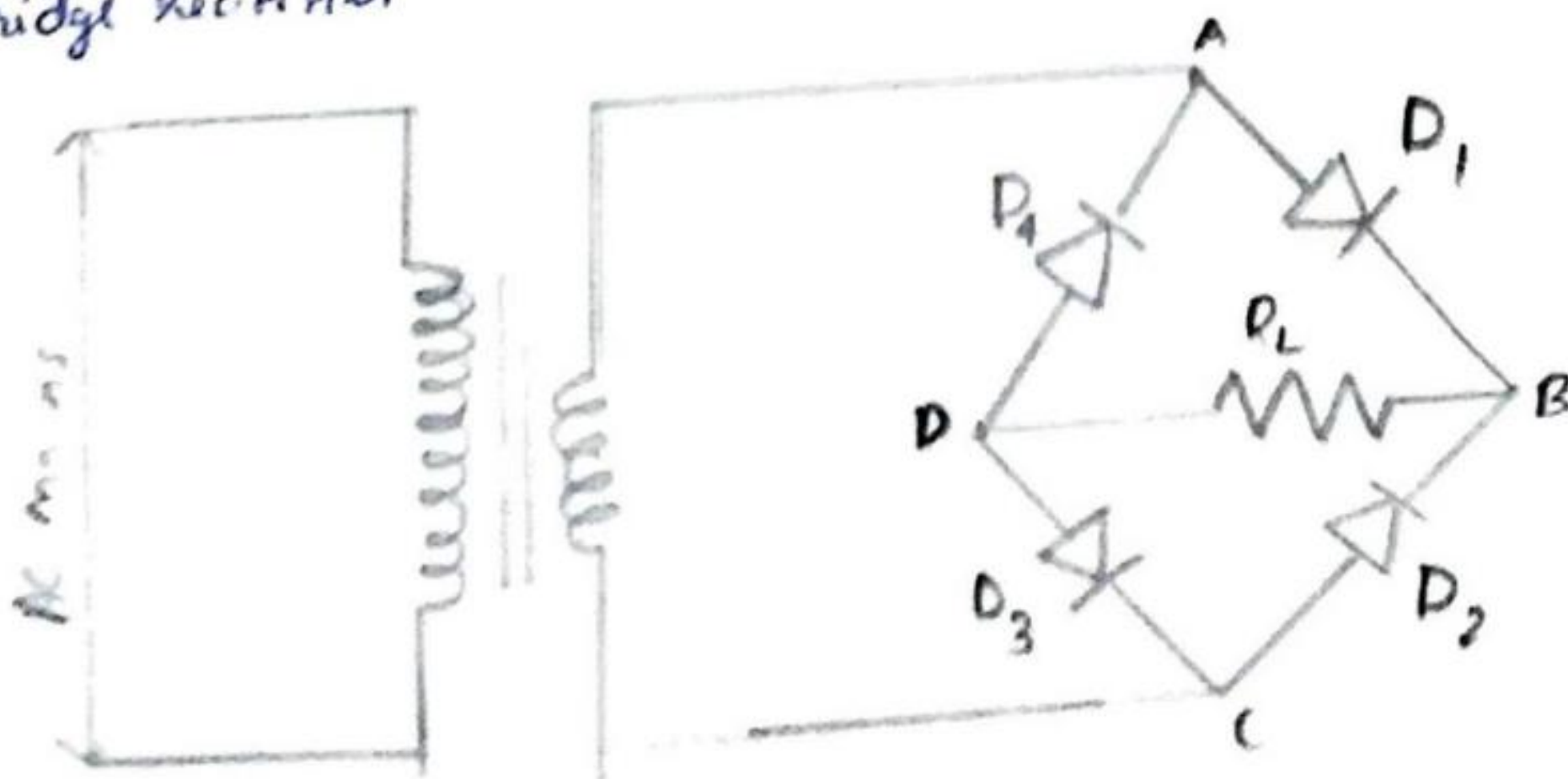
$$\begin{aligned} 18b) \quad \alpha &= \frac{\beta}{\beta + 1} & \beta &= 100 \\ &= \frac{100}{101} \\ &= 0.99 // \end{aligned}$$

19 a)



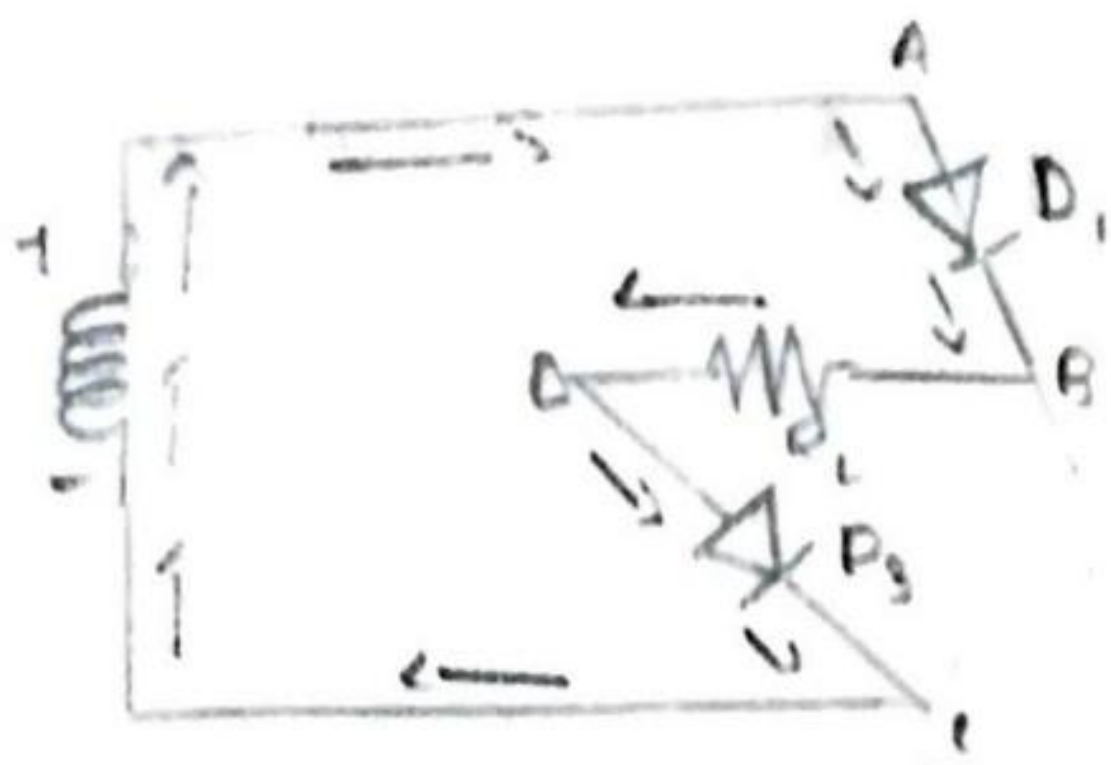
1. Step down transformer :- It step down 220V AC power supply into an AC of required level
2. Rectifier :- Circuit that converts AC into pulsating DC.
3. Smoothing / Filter :- A filtering circuit is used to smoothen the variations present in the rectified circuit.
4. Regulator :- Used to control the unregulated input level to a desired DC output level.
5. Load :- The load is device which is used the pure DC output from the regulated DC output

b) Full wave bridge rectifier



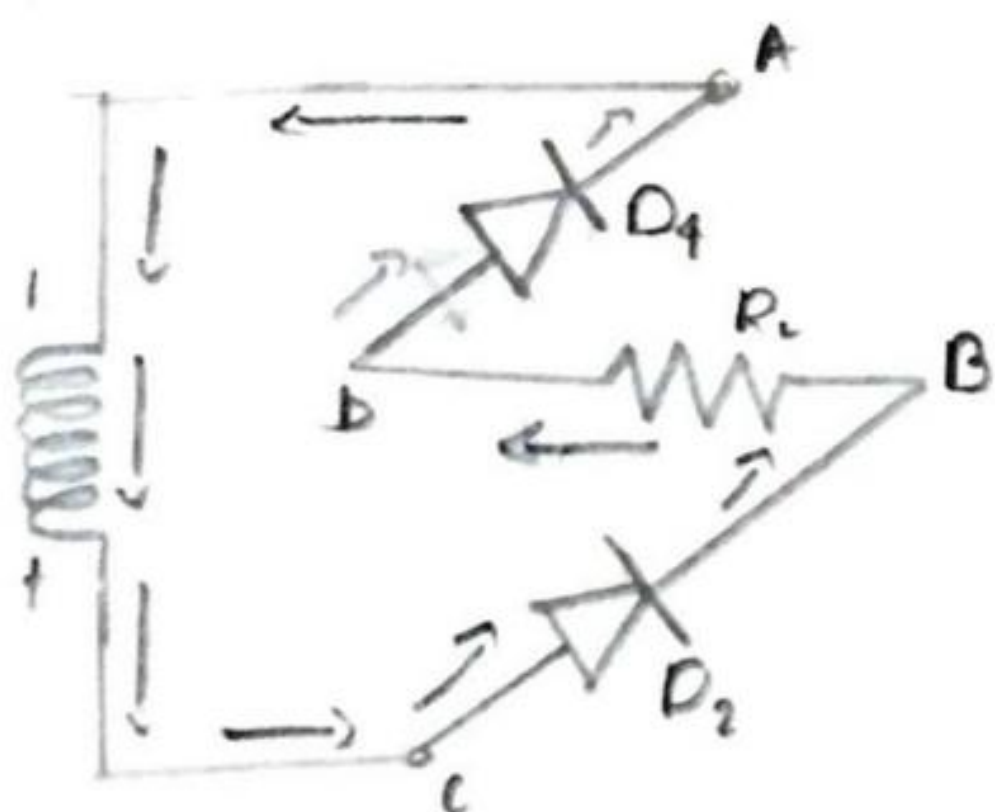
Working:-

Case - 1 :- During +ve half cycle



Here the terminal A is positive with respect to C. Diodes D_1 & D_3 are forward biased. At this instant, the diodes D_2 and D_4 are reverse biased. So the current flows through D_1 , R_L and D_3 .

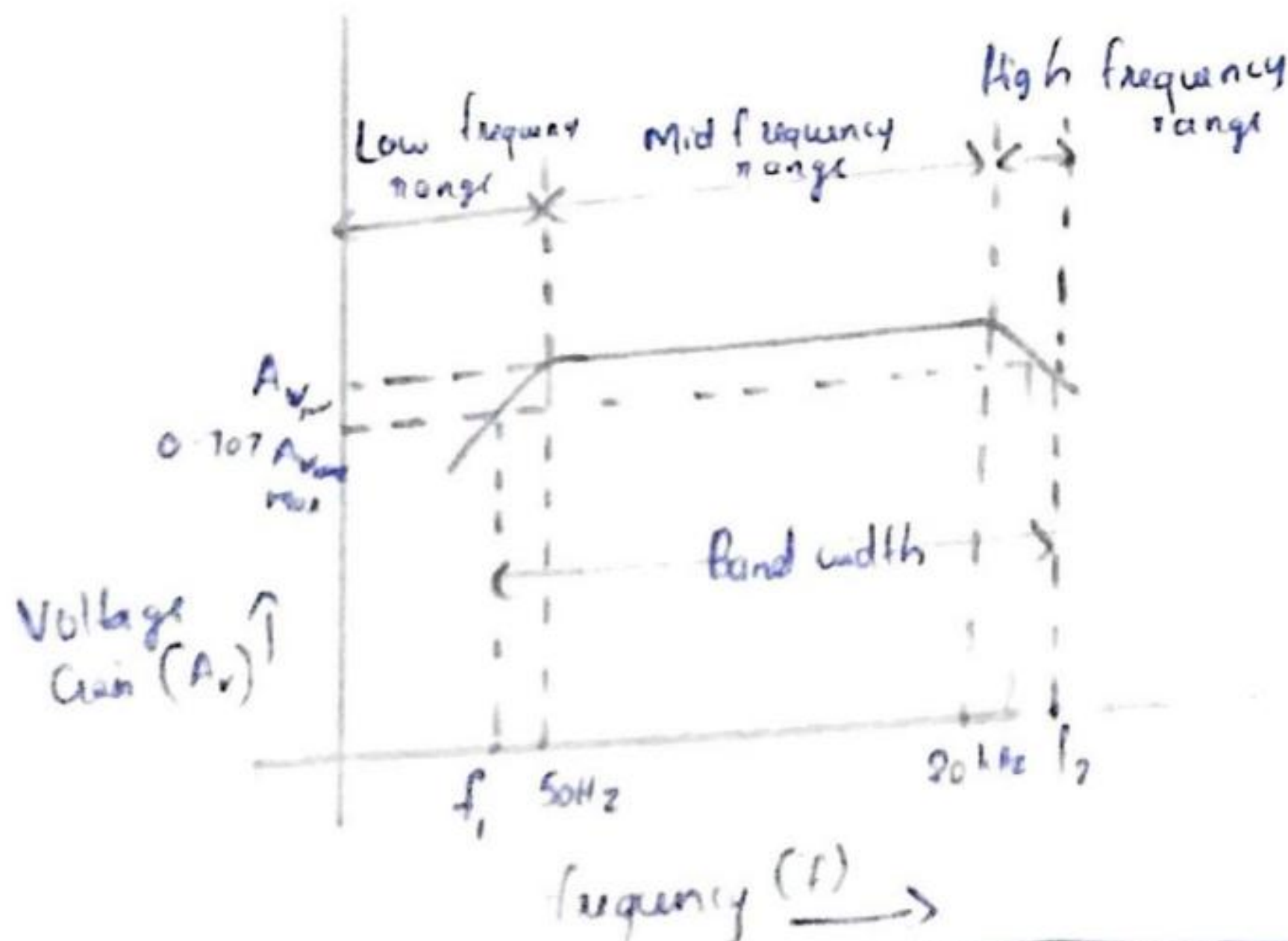
Case-2 :- During -ve half cycle



Here terminal A is negative with respect to C. Diode D_2 & D_4 are forward biased. At this instant D_3 & D_1 are reverse biased. So the current flows through D_2 , R_L and D_4 .

20)

a)



It is evident from the graph that the voltage gain drops off at low frequencies and at high frequencies, while it remains constant in mid frequency.

1) At low frequencies (i.e., below 50 Hz)

- * At low frequencies, capacitor offers a high reactance

 - Coupling capacitors can't effectively couple

 - Emitter bypass can't effectively bypass AC signal

- * As a result, the voltage gain drops off at low frequencies.

2) At high frequencies (i.e., above 20 kHz)

- * At high frequencies, capacitor offers a low reactance

 - Coupling capacitor acts as short circuit

 - Short circuiting effects of junction capacitors of the transistor

- * As a result, voltage gain drops off at high frequencies.

3) At mid frequencies (50 Hz to 20 kHz)

- * Effect of coupling capacitor is such that it maintains a constant voltage gain

- * As the frequency increases, reactance of coupling capacitor decreases, which tends to increase the gain

- * Due to low capacitive reactance, there will be higher loading effect and gain will be reduced.

- * These two factors cancel each other and constant voltage gain is maintained

b) The Emitter by pass capacitor (C_E) is mainly to eliminate its by pass only AC signal to ground. So dc components pass emitter resistor.

This ensure that biasing and stability of transistor is not affected. Without emitter capacitor, the voltage gain of stage would be lost.

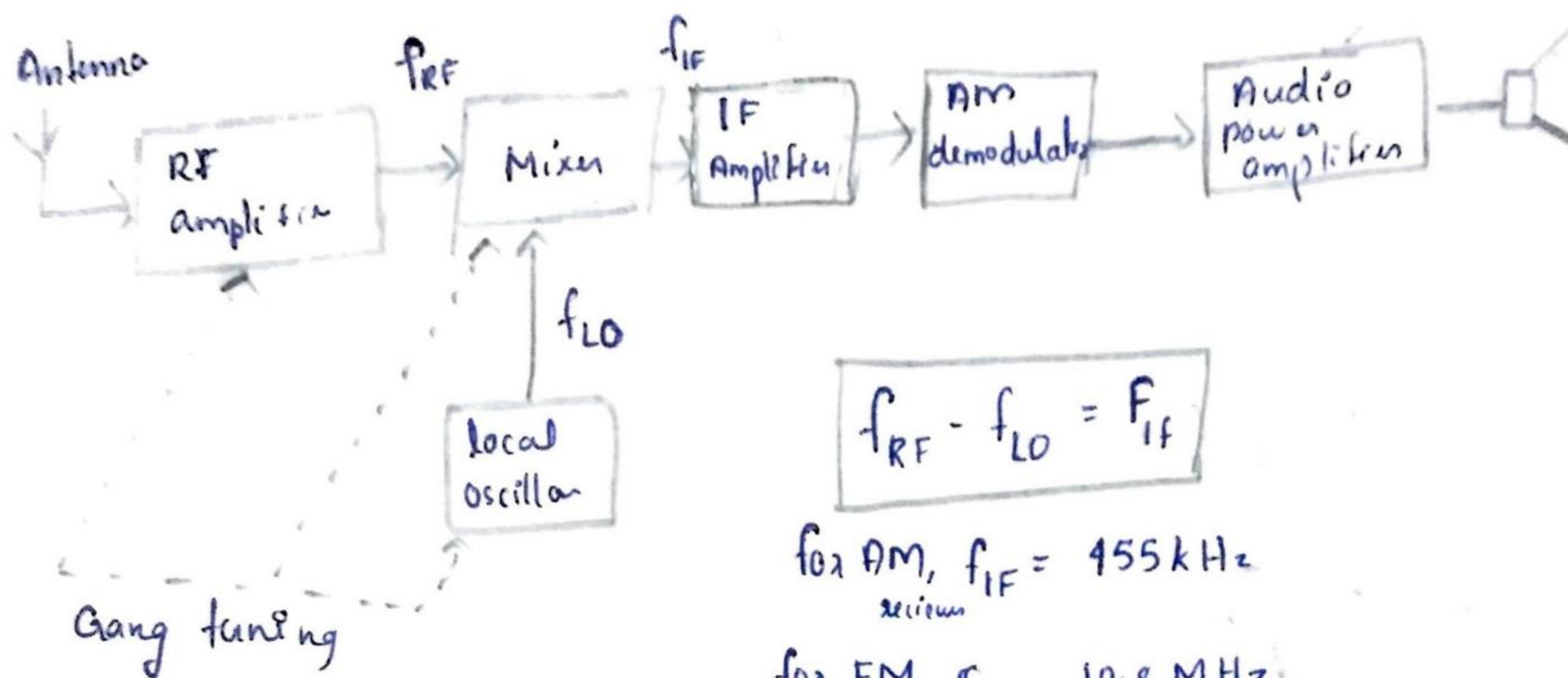
21
a) Advantages:-

- * Easier to implement
- * Demodulation can be done using few components and a circuit
- * The receiver used for AM is very cheap

Disadvantages:-

- * Low efficiency
- * Small operating range
- * Poor audio quality

2.1b) Superhetrodyne receiver



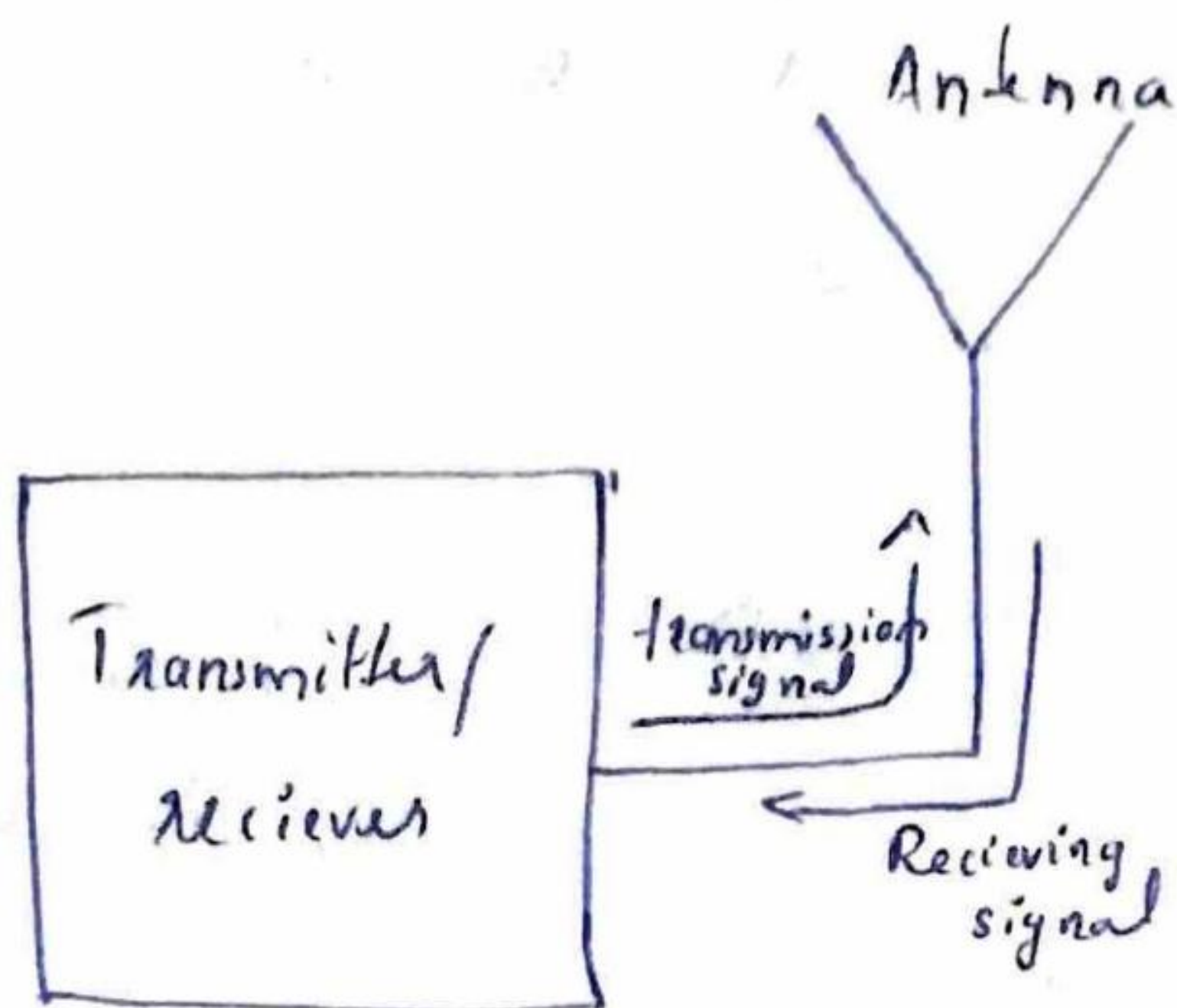
The antenna picks up the signals received from distant broadcasting stations and fed to RF amplifier. RF amplifier selects only the ~~to~~ in decide ~~for~~ (carrier) signal frequency. It rejects all other frequency, also amplifies the ~~we~~ incoming signal.

⇒ The mixer section converts the input to an intermediate frequency (IF) with oscillator.

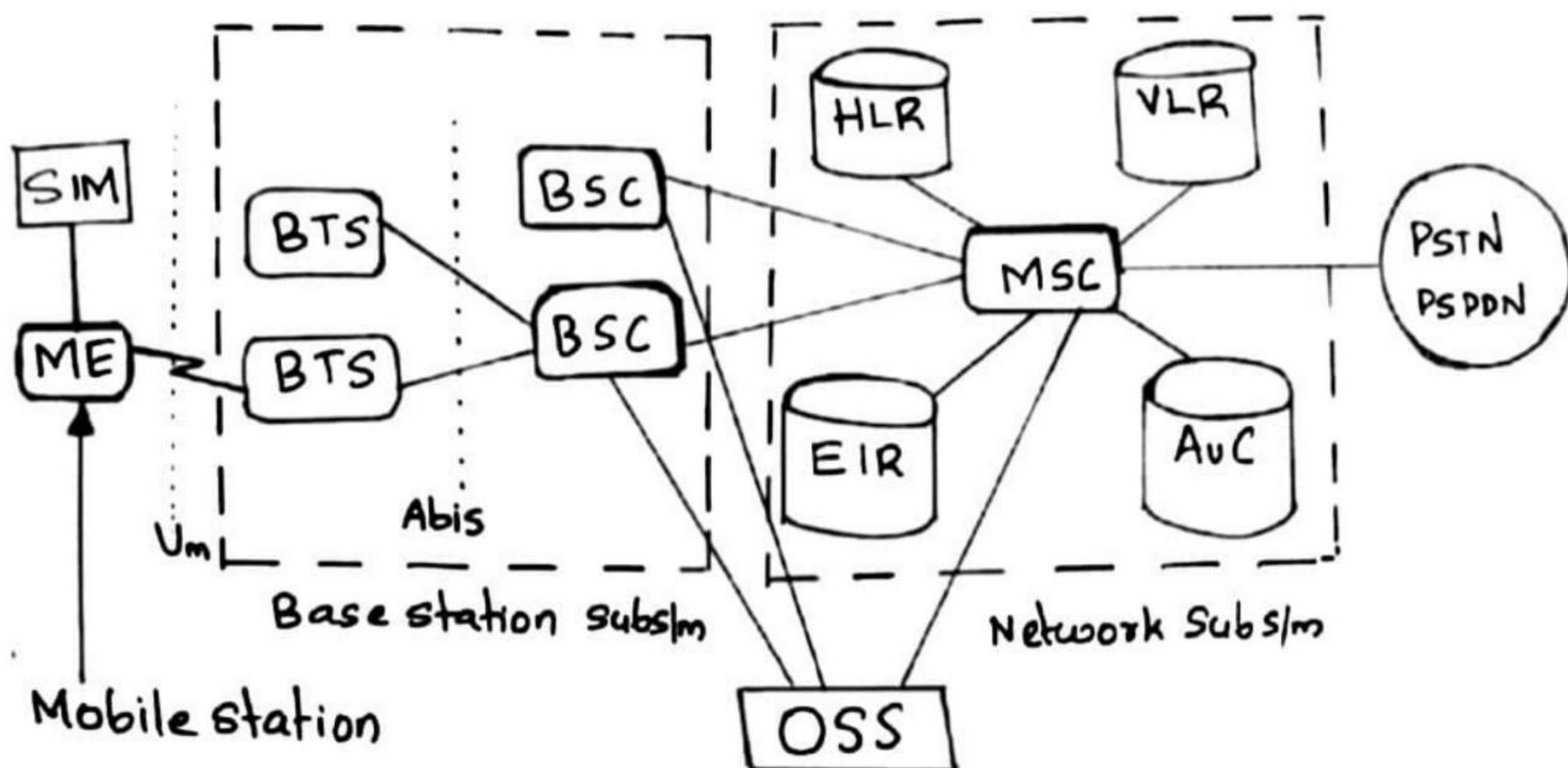
⇒ IF amplifier amplifies the IF signal for

⇒ AM demodulator demodulates the amplified signal and extracts the audio message signal. This audio signal is amplified by an audio power amplifier and is directly given to loud speaker.

22 a)



A stationary charge will create an electric field. If the charge is moving with an acceleration, the electric field will vary. This time-varying electric field produces a magnetic field. The change in magnetic field will also produce an electric field. Thus the mutual interaction of electric and magnetic fields produces an electromagnetic wave.



The mobile station (MS)

→ consists of the physical equipment such as the mobile equipment, and smart card called the SIM (Subscriber Identity Module).

→ SIM provides personal mobility, so that user can have access to all subscribed services irrespective of both the location of the terminal and the use of a specific terminal.

→ By inserting the SIM card into another GSM cellular phone, the user is able to receive calls at that phone, make calls from that phone, or receive other subscribed services.

$$ME + SIM = MS$$

→ The mobile equipment is uniquely identified by the International Mobile Equipment Identity

- The simcard contains the International Subscriber Identity (IMSI) identifying the subscriber, a secret key for authentication, and other user information. The IMEI and the IMSI are independent, thereby providing personal mobility.
- The SIM card may be protected against unauthorized use by a password or personal Identity Number.

Base station Subsystem (BSS)

consists of BTS (Base transceiver station) and BSC (Base station Controller)

- Base transceiver station (BTS) handles the radio interface to the mobile station. The base transceiver station is the radio equipment.
- BTS encodes, encrypts, multiplexes, modulates, and feeds the RF signals to the antenna.
- It communicates with mobile station and BSC.
- Base station controller provides the control functions and physical links between the MSC and BTS. It provides functions such as handover, cell configuration data and control of RF power levels in BTS.

→ A number of BSC are served by
→ BSC manages the radio resource or more BTS. It handles radio setup, frequency hopping and handover.

Network switching Subsystem (NSS)

It consists the following

- a) Mobile switching center (MSC)
- b) Home location Registers (HLR)
- c) Visitor location Registers (VLR)
- d) Authentication Center (AUC)
- e) Equipment identity register (EIR)

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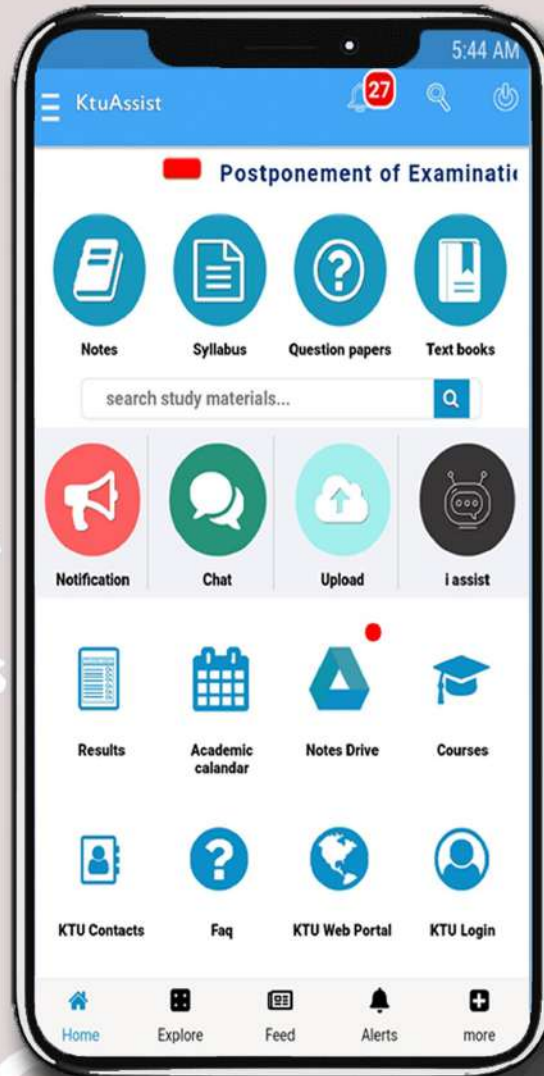
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